

Amrita Vishwa Vidyapeetham
Amrita School of Engineering, Bengaluru
Department of ECE
V Semester B. Tech. ECE and EAC (AY2025-2026)
23ECE381/23EAC381 - Open Laboratory 1

Team ID	TID81	
Domain/Area	VLSI/Embedded/RF/ Signal Processing	
Title of the Project	Customizable Signal Processing Framework with DFT/FFT on TMS320C6x	
Name of the Guide	Dr. Danesh G Kurup	
Students Details	Reg. No.	Name
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Brief Description / Background / Motivation	With increasing demand for real-time signal analysis in modern applications such as radar, autonomous vehicles, and wireless communications, efficient computation of spatial and velocity information is essential. This project aims to implement a two-dimensional Radix-2 Fast Fourier Transform (2D FFT) algorithm on the TMS320C6x digital signal processor for real-time analysis of incoming signal data. The system will take customizable input signals, processes them using DFT in the initial phase and 2D FFT in the advanced phase and the output should identify both position and velocity of moving targets.	
Alignment with UN SDGs	SDG 9, SDG 12	
Objective	To develop a real-time signal processing system on the TMS320C6x DSP that performs Radix-2 2D FFT.	
Sub-objectives (List out all)	1.	Implementing DFT in c.
	2.	Simulating and verifying for TMS board.
	3.	Implementing FFT, 2d FFT on TMS board.
	4.	Plot output in python or MATLAB.
Timeline for the Sub-objectives (Mention Tentative Dates)	1.	July 2025
	2.	Aug 2025
	3.	Sep 2025
	4.	Nov 2025
Software Tools Req'd.	1. MATLAB, Python 2. Code Composer Studios 3. Embedded c	

Hardware Required (Tentative)	1. TMS320C6x https://www.ti.com/tool/TMDSLCDK6748
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